

FlaxSingleWaypointTranslator: Gated Residual Architecture ($s_t, B(w_1) \rightarrow z_{\text{cmd}}$)

Current State $s_t \in \mathbb{R}^{d_s}$

Continuous Agent State ($d_s = 29$)

1a State Feature Extraction

Dense(256) $\rightarrow$ LN $\rightarrow$ GELU $\rightarrow$ Dense(256) $\rightarrow$ LN
 $h_s \in \mathbb{R}^{256}$ (Kinematic & Spatial)

Lookahead Waypoint $B(w_1) \in \mathbb{S}^{d-1}$

Immediate Dijkstra Target ($d_z = 128$)

1b Waypoint Feature Extraction

Dense(256) $\rightarrow$ LN $\rightarrow$ GELU $\rightarrow$ Dense(256)
 $h_w \in \mathbb{R}^{256}$ (Target Corridor)

2 State-Adaptive Gating Mechanism

Concatenation: $[h_s, h_w] \in \mathbb{R}^{512} \Rightarrow \gamma = \sigma(\text{Dense}([h_s, h_w])) \in (0, 1)^{256}$
Convex Gated Combination: $h_{\text{fused}} = \gamma \odot h_w + (1 - \gamma) \odot h_s \in \mathbb{R}^{256}$

Self-modulates waypoint pull based on current obstacle proximity and momentum

3 Deep Residual Refinement Blocks ($N_{\text{layers}} = 3$)

$h^{(l+1)} = h^{(l)} + \text{GELU}\left(\text{LN}\left(\text{Dense}_{256}\left(h^{(l)}\right)\right)\right), \quad l \in \{1, \dots, N-1\}$

Dense + LN + GELU $\rightarrow$ $+$

4 Spherical Head $\mathbb{S}^{d-1}$

Dense(128) $\rightarrow \hat{z}$
 $z_{\text{cmd}} = \sqrt{d} \cdot \frac{\hat{z}}{\|\hat{z}\|_{2+10^{-8}}} \in \mathbb{S}^{d-1}$

5 Policy Execution π^ℓ

Action Output: $a_t = \text{mode } \pi^\ell(s_t, z_{\text{cmd}})$
 $\mathcal{A} \in [-1, 1]^8$ (Ant Joint Torques)

Direct $s_t \rightarrow w_1$ **Coupling:** $O(1)$ fast

Adaptive Gate γ : Smooth transition

Radius Preservation: $\|z_{\text{cmd}}\|_2 = \sqrt{128}$